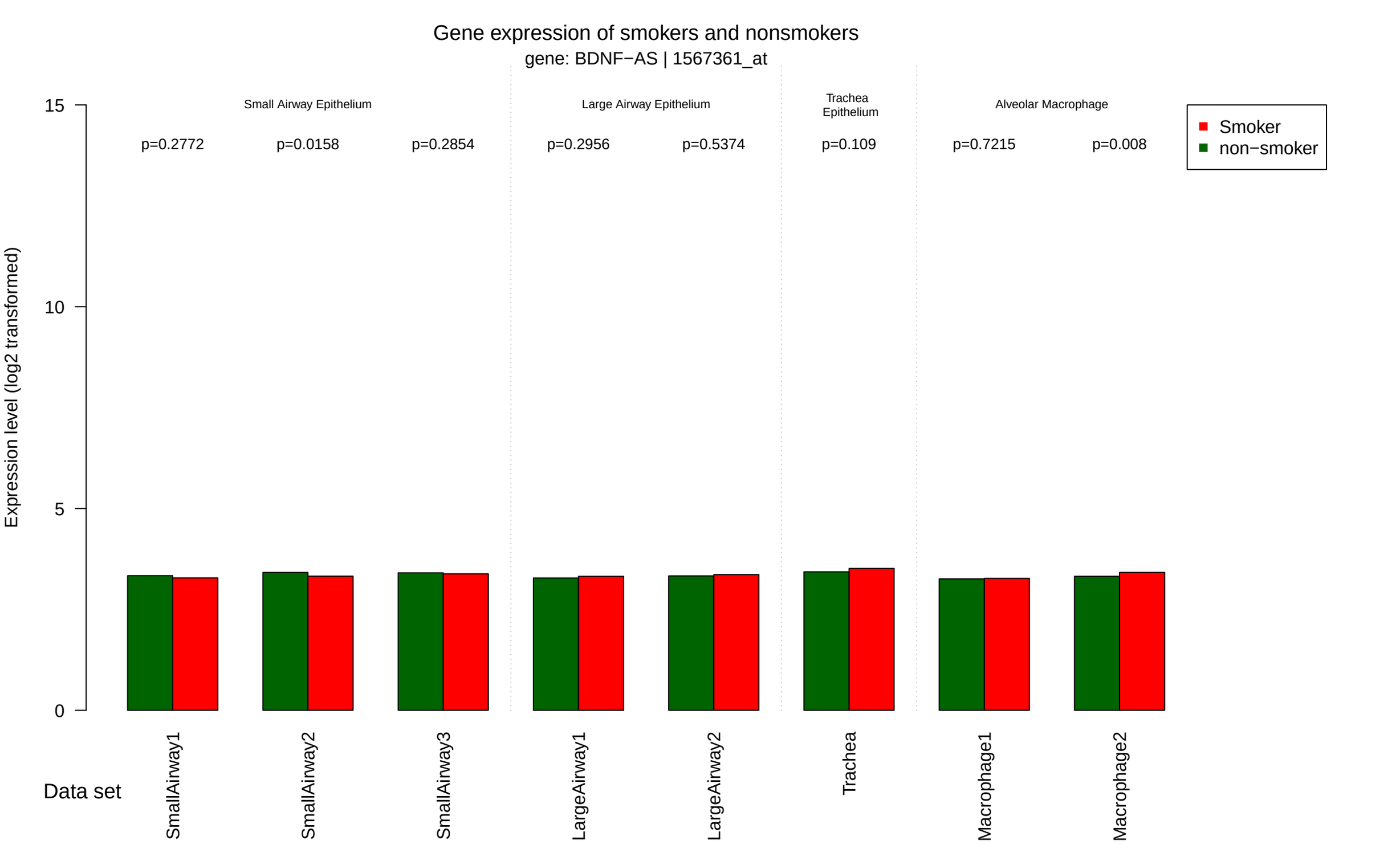
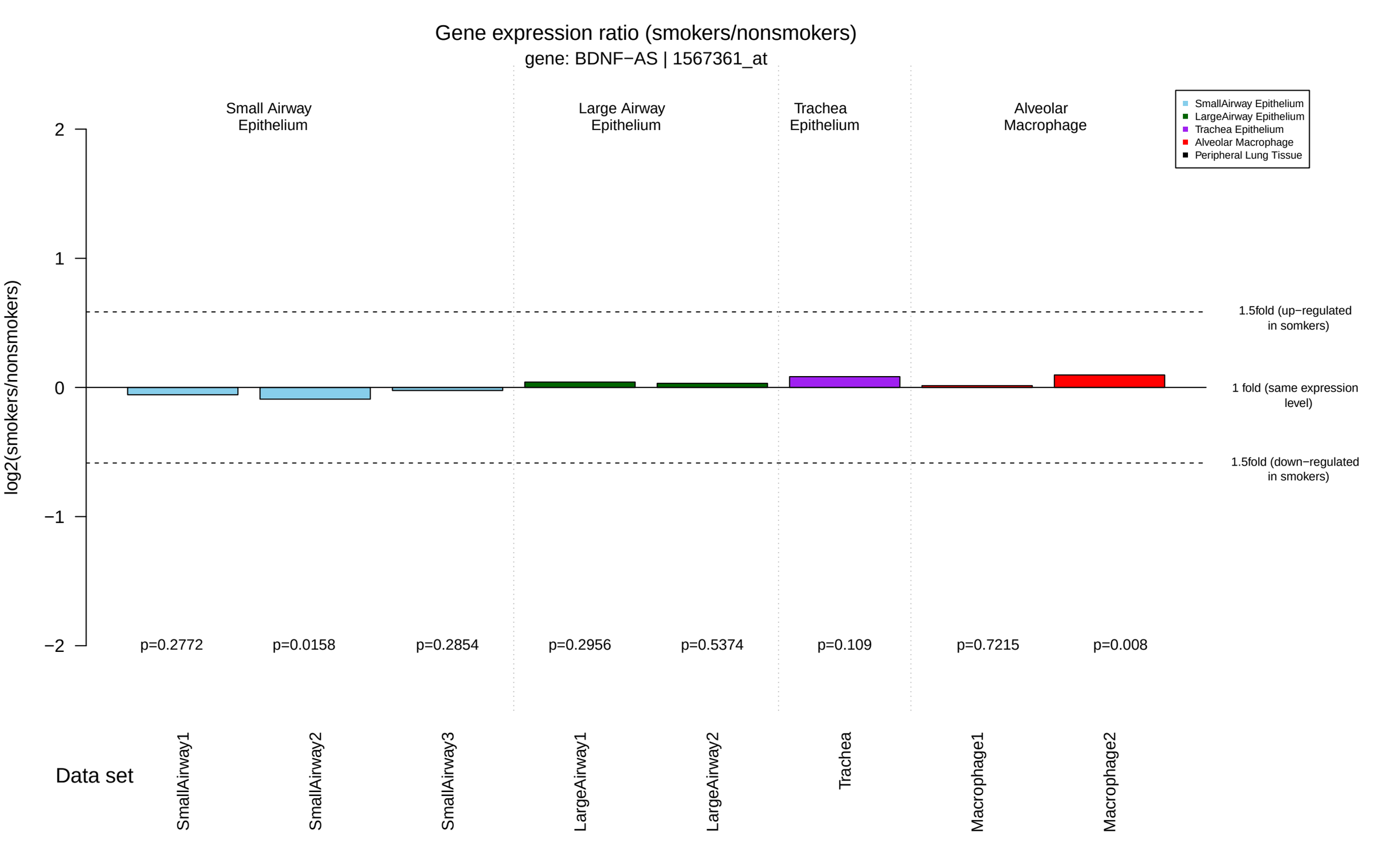
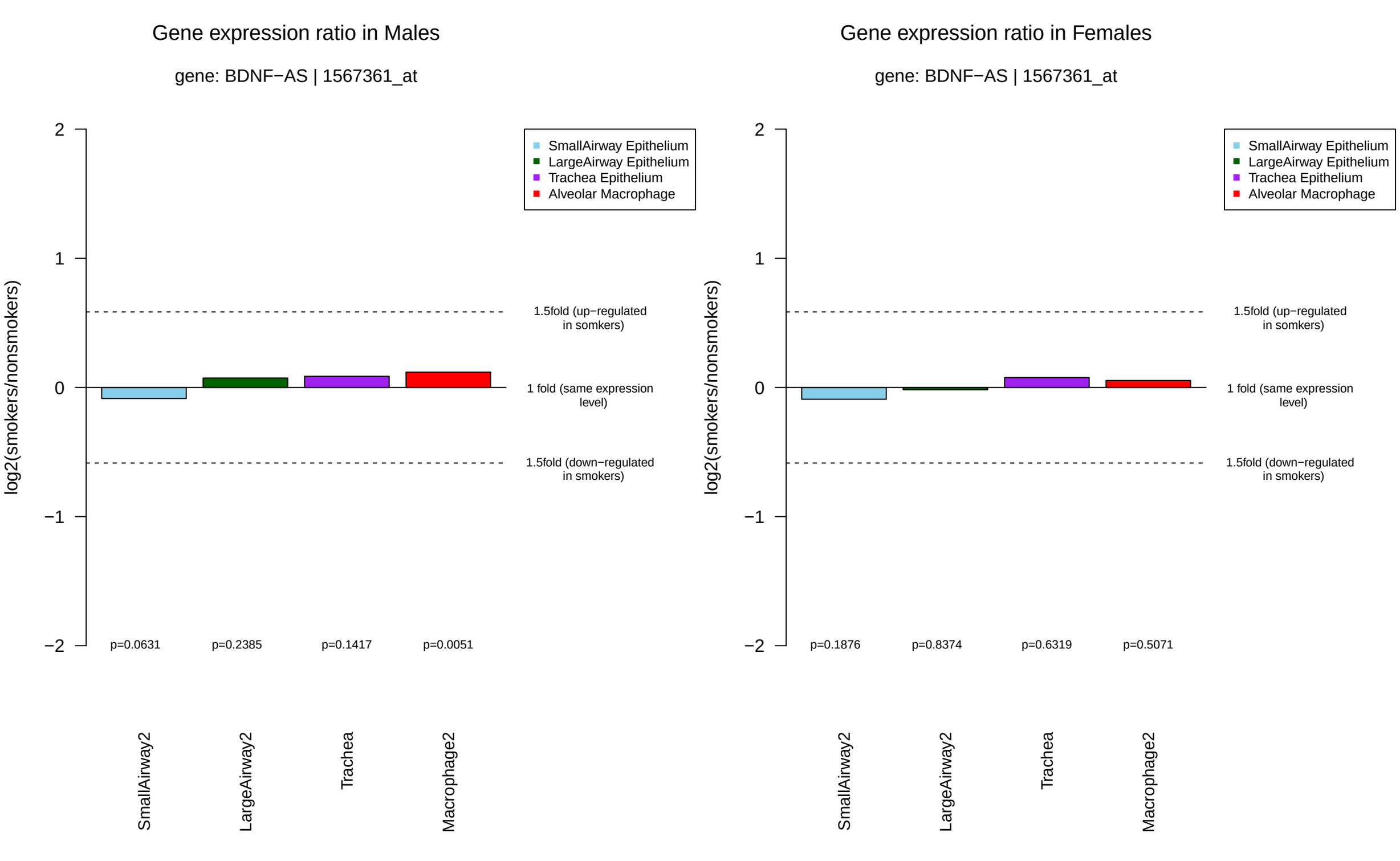
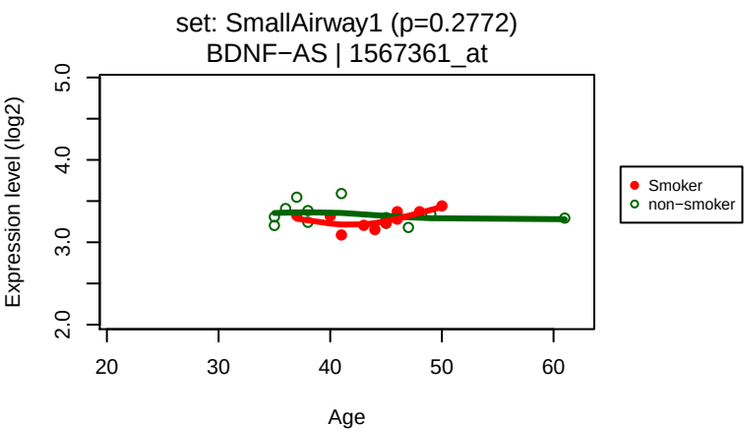
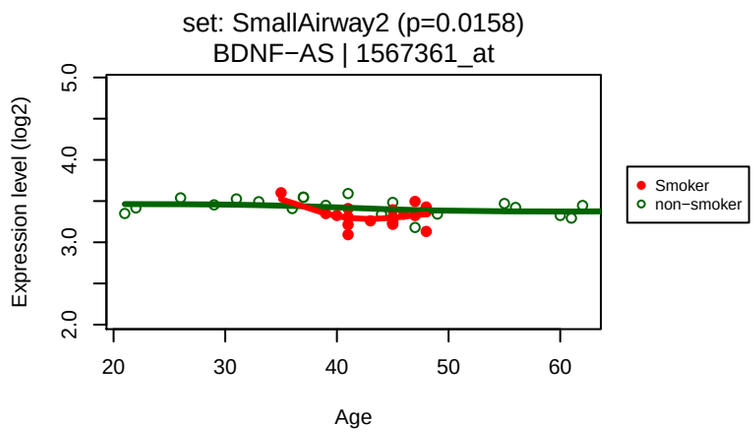
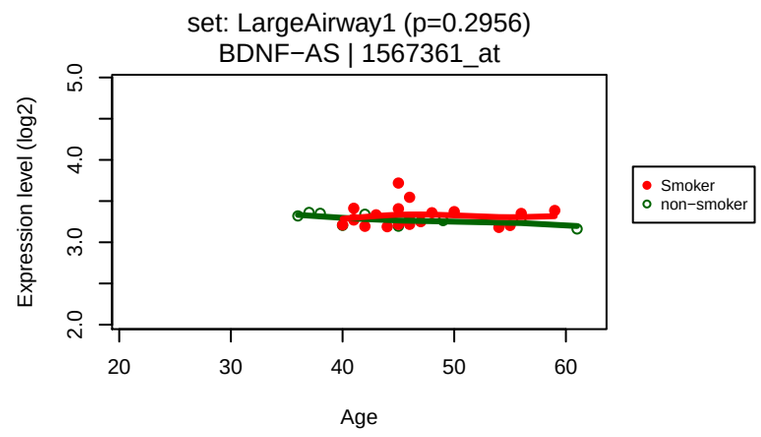

Gene expression of smokers and nonsmokers
gene: BDNF−AS | 1567361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2772
p=0.0158
p=0.2854
p=0.2956
p=0.5374
p=0.109
p=0.7215
p=0.008
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

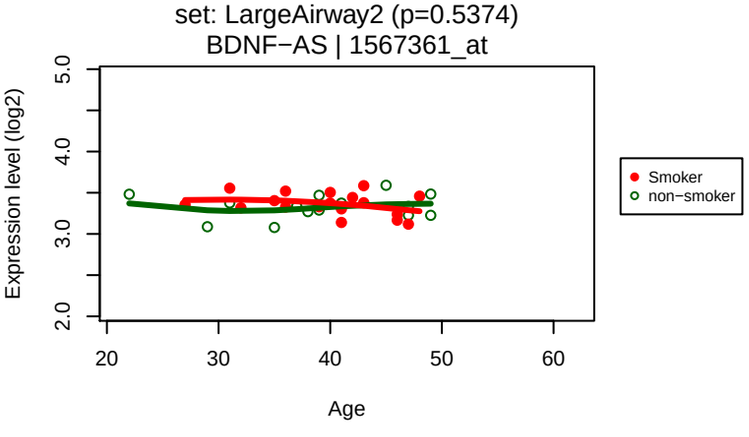

Gene expression ratio (smokers/nonsmokers)
gene: BDNF−AS | 1567361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2772
p=0.0158
p=0.2854
p=0.2956
p=0.5374
p=0.109
p=0.7215
p=0.008
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

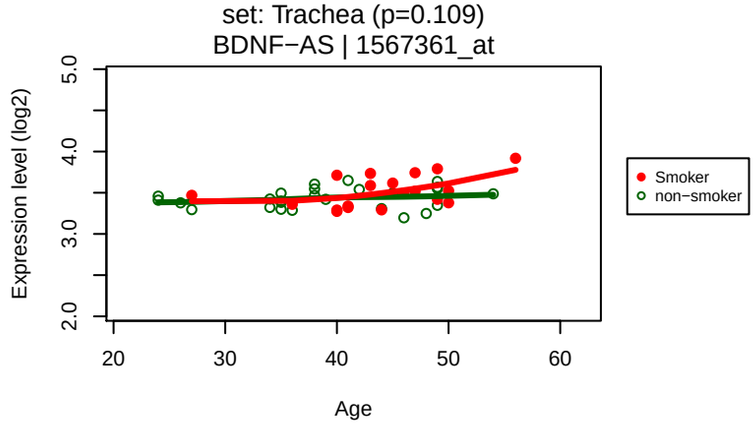

Gene expression ratio in Males
gene: BDNF−AS | 1567361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0631
p=0.2385
p=0.1417
p=0.0051
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BDNF−AS | 1567361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1876
p=0.8374
p=0.6319
p=0.5071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

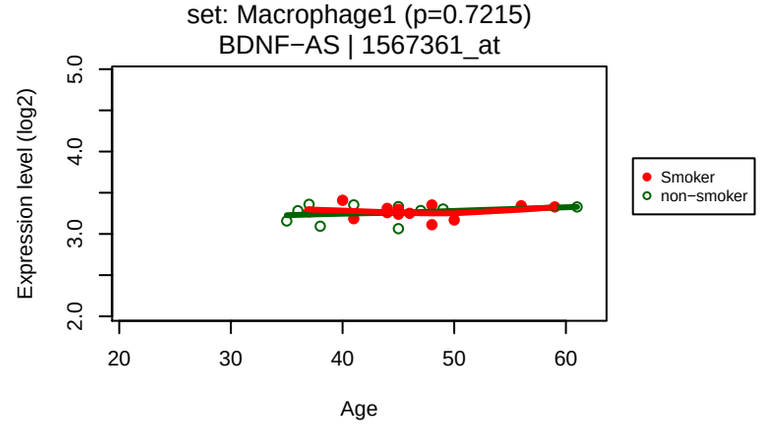

set: SmallAirway1 (p=0.2772)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker

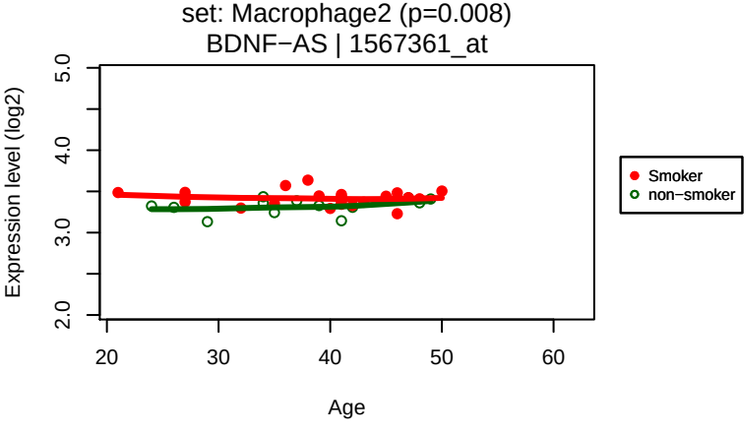

set: SmallAirway2 (p=0.0158)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker

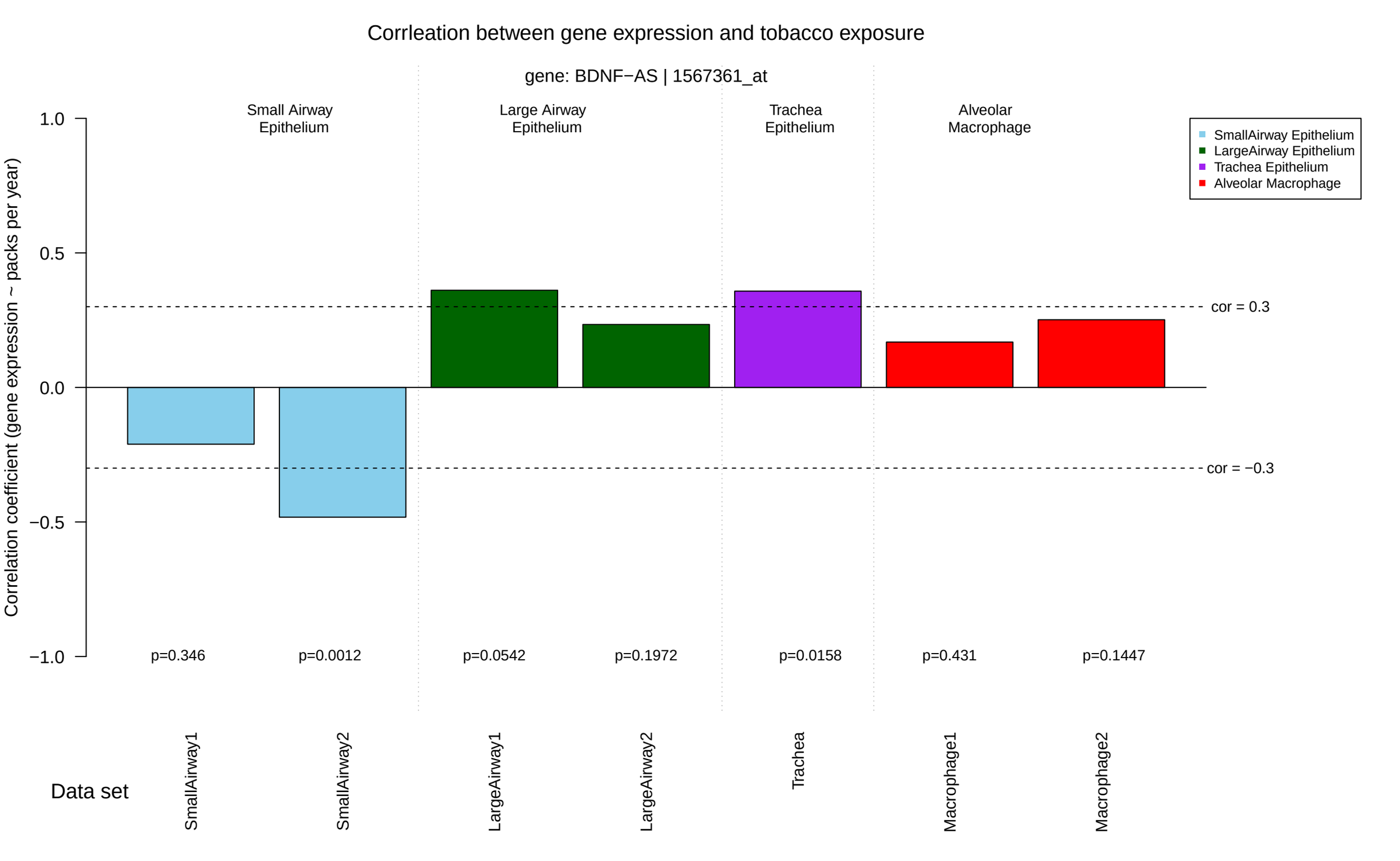

set: LargeAirway1 (p=0.2956)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5374)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.109)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7215)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.008)
BDNF−AS | 1567361_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BDNF−AS | 1567361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.346
p=0.0012
p=0.0542
p=0.1972
p=0.0158
p=0.431
p=0.1447
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2